

PRD: H-1B Registration Readiness and Control Layer

TL;DR

- The problem: USCIS already automates the portal. The real work is everything it does not, which is checking that data is substantively correct, deduplicated, and optimally filed, during the months before the window when USCIS gives zero feedback, on behalf of an attorney who is personally liable.
- What we automate: data intake and normalization, substance and format validation, duplicate detection, USCIS template generation, status and audit capture.
- How: a readiness layer on Alma's existing case graph, organized around one primitive, the readiness ledger, with AI assisting the judgment fields and grounding every suggestion in a source.
- What we keep manual: final sign-off, ambiguous determinations and duplicates, the G-28 signature, and payment authorization.
- Why: in a high-stakes, irreversible, deadline-bound workflow, the unit of value is a defensible safe to submit decision, not an automated click. Maximize safe throughput, not automation percentage.
- Prototype: <https://h1b-console.vercel.app>
- GitHub Repo: <https://github.com/sukeerthiii/Alma-H-1B-Lottery-Registration>

2. Scope

- 📌 USCIS already automates the portal mechanics. The video shows bulk upload of up to 2,500 beneficiaries, automatic draft creation, batching into groups of 250, a duplicate checker, a payment handoff, and confirmation capture. Re-automating a portal that already accepts a spreadsheet is low-leverage and brittle.

The scope falls out of four facts, deductively.

- **USCIS validates format, never substance.** Its error demo flags a malformed wage level and an invalid country name. A wage level that is well-formatted but legally wrong, a valid-but-incorrect country, a mis-chosen SOC code, or a duplicate person all pass silently.
- **USCIS dedup is optional and blind.** It is opt-in, compares only against already-submitted registrations, and explicitly does not check within a draft, between drafts, or across organizations.
- **The window is a gate, not a workspace.** Data entry is locked until the period opens (FY2027: noon ET March 4 to noon ET March 19, 2026). Preparation happens over the prior months with zero USCIS feedback.
- **The attorney is accountable.** The representative signs the G-28 and is professionally liable.

Therefore the product owns everything USCIS will not, during the months USCIS gives no signal, on behalf of the person who carries the liability: data substance, deduplication, determinations, and a defensible record of why a batch was safe to submit.

3. The Problem, Sized

H-1B registration is a high-volume, deadline-bound legal-operations workflow with a structural blind spot.

The volume math: a single mid-market client can bring dozens to low hundreds of beneficiaries. An attorney or pod handles many clients at once. USCIS itself estimates 56 minutes of work per registration. At a few hundred registrations, that is hundreds of hours, compressed against a fifteen-day wall that opens once a year.

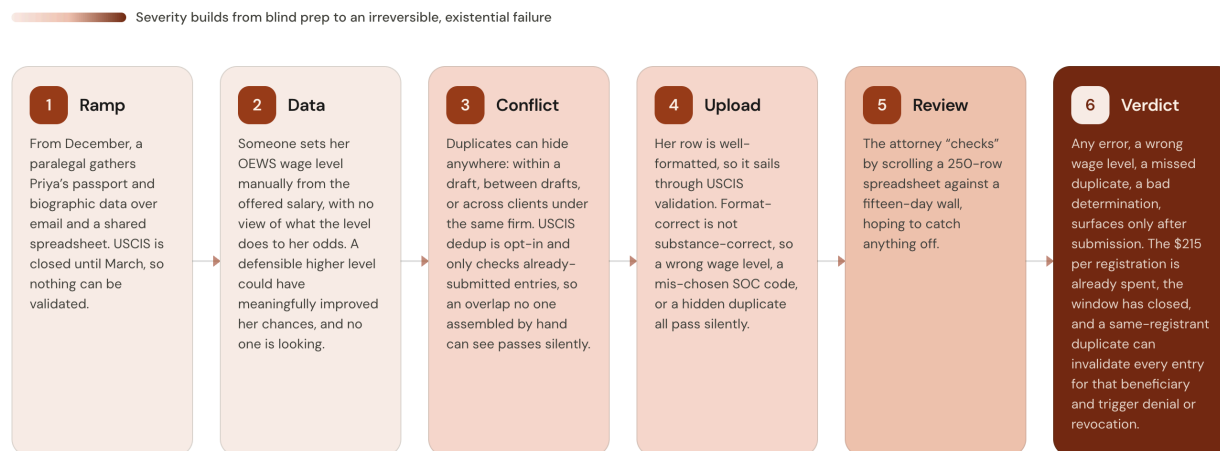
The structural leaks:

- For months before the window, attorneys and paralegals assemble data, make wage and occupation determinations, and try to prevent duplicates with no validation surface, because USCIS is closed until March.
- USCIS, the only system that will eventually check the data, checks the shallow layer (format) and punts the deep layer (substance, dedup) back to the attorney.
- Then, in two weeks, the attorney certifies hundreds of irreversible registrations across many clients. Each is a non-refundable \$215, and a single same-registrant duplicate invalidates all of that registrant's entries for the beneficiary and can trigger denial or revocation.
- Selection is now weighted by wage level, so the wage determination is no longer just compliance. It is a direct lever on the client's odds.

The bottleneck is trust at volume. The attorney has no trustworthy way to know, across hundreds of beneficiaries and many clients, which registrations are correct, non-duplicate, and optimally filed.

4. One Beneficiary Today, End To End

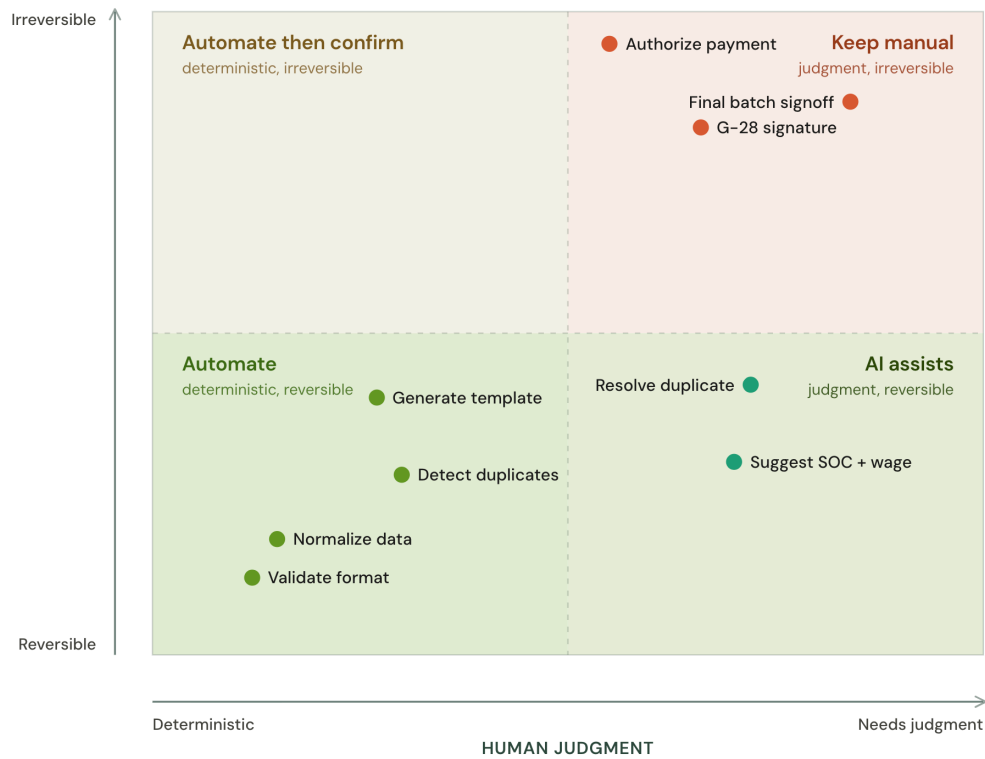
To make the above concrete, follow Priya Nair, a software engineer at a client called Northwind.



Every capability in this PRD exists to make that path fast and safe for hundreds of Priyas at once.

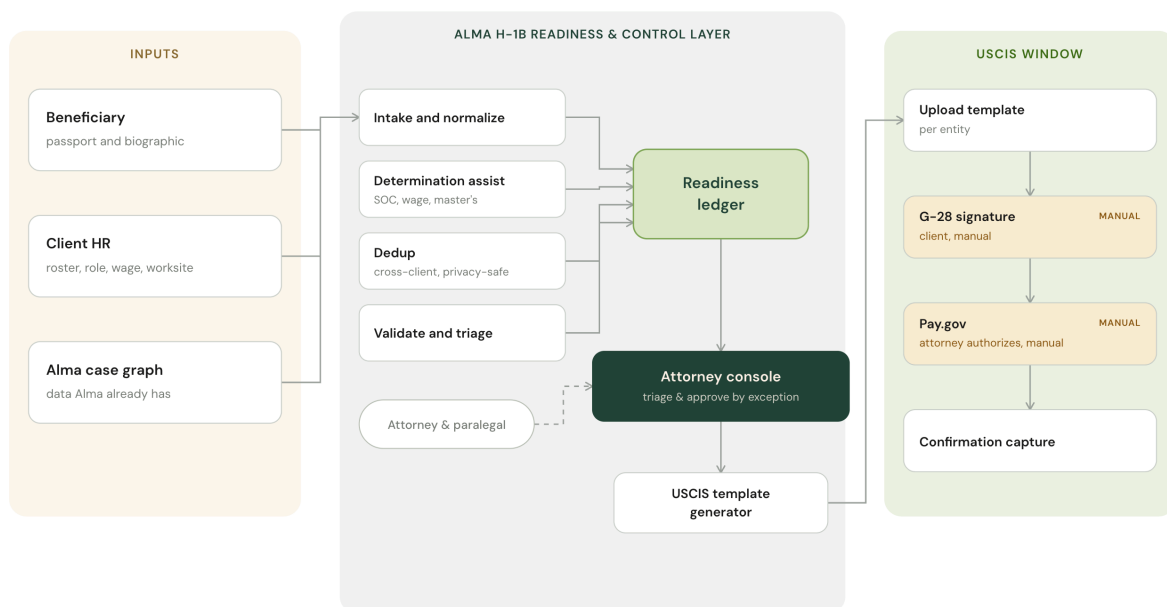
5. What We Automate, How, What We Keep Manual, And Why

The logic: place every task by two axes, whether it is reversible and whether it needs human judgment. That placement decides **automate, assist, or keep manual**.



6. The Solution

The one-line model: the product is a layer that sits between Alma's data and the USCIS window. Inputs flow into a set of engines that all feed one object, the readiness ledger, which drives the attorney console, which emits the USCIS template.



Read it in three phases: the product sits on top of data Alma already has, four engines feed one ledger, and the two human gates (signature, payment) live downstream in the USCIS window.

7. The Readiness Ledger

Every registration carries a structured list of checks, each with a state and a source. The attorney approves by trusting the ledger and reviewing the validators. This is both the trust mechanism and the throughput unlock.

The engines feeding the ledger: intake and extraction, normalization, determination assist, dedup, validation and readiness, template generation, and submission assist with confirmation capture.

Priya Nair

Ready Austin, TX

All checks pass. Ready for approval.

READINESS LEDGER

✓ Identity

Passport AB1234567 (India)

Passport scan

✓ Biographic

India · born 1996-04-12

HRIS

✓ SOC code

15-1252 Software Developers

Attorney

✓ Wage level

Level II

AI 93% · OEWS

✓ Duplicate

No unresolved match in firm

Firm-wide dedup

☐ G-28 signature

Pending client admin

Client

BENEFICIARY DATA

Edit all fields

Pre-filled from Alma's data. You can override every field that goes to USCIS.

WAGE LEVEL · ODDS DIAL

AI suggests **Level II** · 93% confidence. Offered wage equals or exceeds the OEWS Level II threshold for this SOC in Austin, TX. (DOL OEWS, snapshot at registration)

LEVEL I

1×

\$98,000+

LEVEL II

2×

\$122,000+

LEVEL III

3×

\$148,000+

LEVEL IV

4×

\$178,000+

Offered wage \$128,000 supports up to **Level II** (2 lottery entries).

CAPTURED FOR THE PETITION

Min degree: Bachelor's in Computer Science

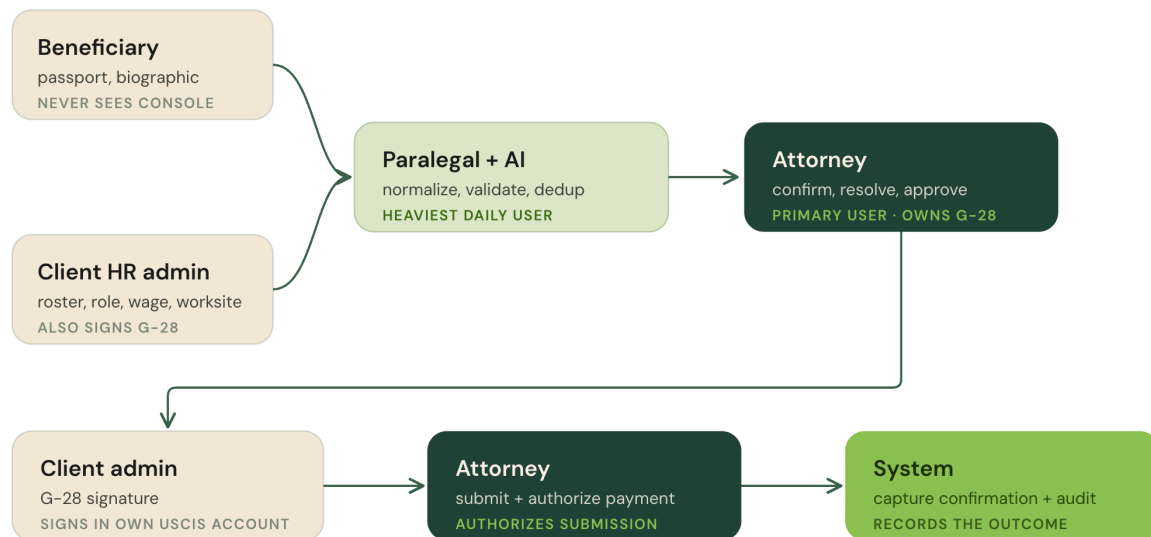
Experience: 36 months · supervises 0

Wage basis: \$128,000, OEWS snapshot dated at registration

These feed the LCA and the I-129, where wage level, SOC, and worksite must stay consistent.

8. Users and Handoffs

The interface we build is the attorney's, but it sits on top of work others start.



- **Beneficiary:** source of truth for passport and biographic data. Needs low-friction, confidence-building intake. Never sees the console.
- **Client HR admin:** provides roster and role, wage, and worksite context, and legally signs the G-28 in their own USCIS account.
- **Paralegal or legal ops:** the heaviest day-to-day user, clears corrections and preps batches.
- **Attorney:** owns judgment, confirms determinations, resolves duplicate risk, and authorizes submission. Primary user because the prompt targets the attorney interface and because the attorney carries the liability. The paralegal works the same surface in a correction mode.

9. The Attorney Console

The prototype is the live attorney console at <https://h1b-console.vercel.app>.

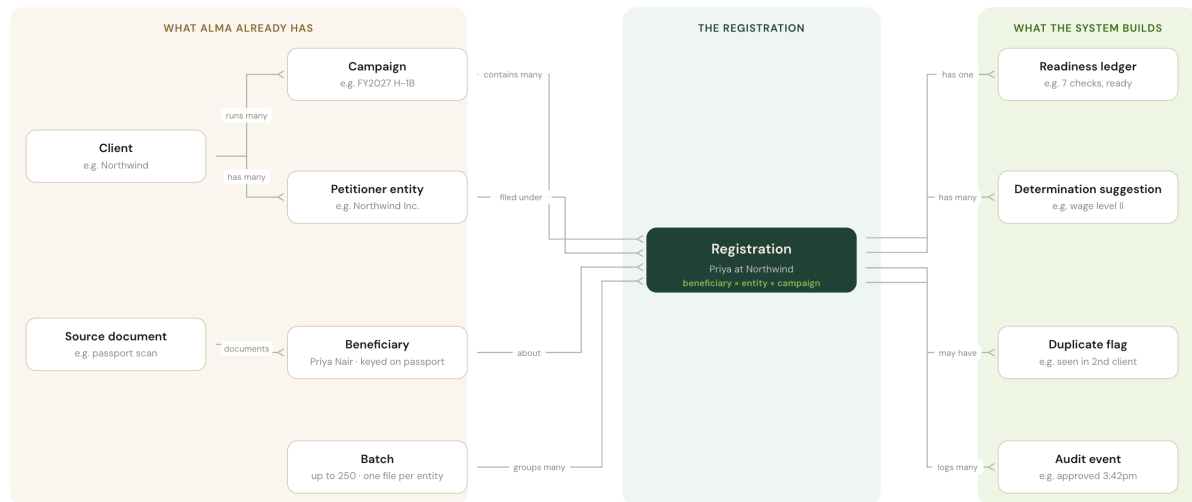
📌 How it works in detail: <https://h1b-console.vercel.app/how-it-works>

- **Firm portfolio:** every client campaign in one view with readiness at a glance. The firm-wide totals (beneficiaries, ready, exceptions) open a cross-client All beneficiaries table, so an attorney can work exceptions across clients, not just one campaign. This cross-client view is only possible because Alma holds the whole book in one graph.
- **Campaign workspace:** the readiness funnel (clickable status filters), a petitioner-entity filter (USCIS files are per entity), a beneficiary table, and one-action Approve clean set.
- **Exception triage drawer:** the readiness ledger; full manual editing of every USCIS field, pre-filled by automation but fully overridable so the attorney keeps ultimate control; the cross-client duplicate with a privacy-safe ethical wall; and the wage-level odds dial. (shown in the prev. section)
- **Window-open mode:** the batch submission monitor, per-entity USCIS template export (CSV in the exact template columns, one file per petitioner entity), payment and signature as manual gates, confirmation capture, and a selection-to-petition handoff.

- Audit log: every action recorded per campaign, who and when.

10. Data Model

The design fact that powers the differentiators: the beneficiary is keyed on passport and lives above any single client, while a registration is the intersection of beneficiary, petitioner entity, and campaign. Batches respect USCIS limits (up to 250 per batch, separate files per entity). The beneficiary-centric rule plus Alma's single-roof data make firm-wide dedup both necessary and uniquely possible.



📌 Registration as the on-ramp to the petition

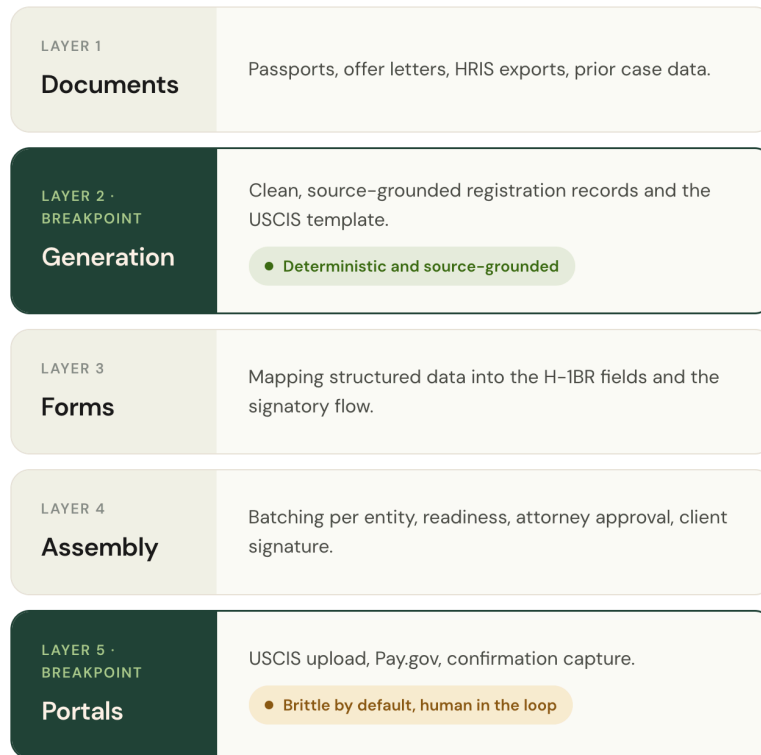
The registration is not a dead end. For FY2027, USCIS ships a new I-129 that cross-references the registration: the wage level, SOC, and worksite chosen at registration must stay consistent through the certified LCA and the petition, and a downgrade triggers an RFE or denial. So the determinations made here are the spine of the petition.

11. Two Flywheels

- **Cross-client dedup gets safer with every client.** More of the firm in one graph means a more complete duplicate and conflict signal, which makes filings safer, which earns more clients.
- **The determination engine gets smarter every season.** Each cycle adds confirmed SOC and wage-level calls across clients, which improves next season's suggestions. AI leverage that compounds, specific to Alma's position as the firm.

12. Map To Alma's Five-Layer Model

The two internal breakpoints (source: call with Abhinav), **document generation** and **web automation**, are exactly where this product concentrates care:



- 📌 A natural extension is a legal query agent over this data ("which registrations are blocked, which have duplicate risk, which are missing passports"), grounded in the same ledger and audit log.

13. Metrics

Measuring safe throughput here over automation percentage. Targets are directional, to be calibrated against a baseline season.

METRIC	TODAY	TARGET DIRECTION
Intake to ready batch Time from roster intake to a ready batch	Long, manual assembly over months	→ Shorter, engine-driven prep
Attorney review per registration Review time per clean registration	Per-row eyeballing of the spreadsheet	→ One bulk approval per batch
Auto-validated before review Percent of registrations auto-validated before attorney review	Low: little validated up front	→ High: most arrive pre-checked
Campaign ready before window Percent of the campaign ready before the window opens	Scramble during the open window	→ Most of it ready before it opens
Risks caught before window Duplicate and conflict risks caught before the window	Found post-submission, too late	→ Zero discovered post-submission
Rework and confirmation Rework rate after approval, confirmation-capture rate	Silent failures, gaps in the trail	→ Low rework, full confirmation capture

Vanity metrics to avoid: automated clicks, AI-generated fields, percent “fully automated.”

14. Potential Risks

- USCIS changes the template or portal. The template generator is the only USCIS-coupled component, isolated and versioned. (easy fix)
- Source data is wrong. Validate at intake, let beneficiaries self-confirm.
- Attorney over-trusts the automation. The ledger shows what was and was not checked, with confidence always visible.
- Client admin signature stalls near the deadline. Status tracking, reminders, escalation, decline-reason capture.
- Payment fails to clear after a submitted state. Payment readiness checks, card limit and ACH guidance, reconciliation against submitted registrations.

15. Anticipated Questions

- **Confidentiality and conflicts on cross client dedup.** A firm already owes conflict checking, so within firm dedup is defensible. We surface only a privacy safe passport hash signal, never the other client's identity, behind an ethical wall and a conflicts review.
- **Is dedup still material under beneficiary-centric selection.** Yes. A same-registrant duplicate invalidates all of that registrant's entries and risks denial. A same-person-across-two-clients case is a conflict event, not a mere duplicate.

- **Does the wage dial encourage gaming.** No. The system never changes the offered wage. Any higher level must reflect a genuinely higher bona fide offer that the employer and attorney decide.
- **Can AI be trusted with SOC and wage.** It assists, it never decides. Grounded in DOL and OEWS data, with rationale, confidence, and source, low confidence routes to the attorney, and the attorney confirms every determination.
- **PII and security.** Passports and dates of birth are sensitive, so encryption, role-based access, and a full audit trail are first-class.

16. Out of scope

Selection (by late March) and the downstream I-129 petition (April to June) are out of scope here. The registration data this product produces feeds the petition cleanly, which is the natural next surface.